

RIP & EIGRP

- Topics covered:
 - Routing Information Protocol (RIP)
 - Enhanced Interior Gateway Routing Protocol (EIGRP)
-

Routing Information Protocol (RIP)

Review of RIP Basics

RIP stands for Routing Information Protocol.

It is an industry standard protocol, not Cisco proprietary.

RIP is a distance vector interior gateway protocol.

It uses routing-by-rumour logic to learn and share routes.

RIP uses hop count as its metric.

Each router in the path counts as one hop.

- Bandwidth is irrelevant
- 10 Gbps = 1 hop
- 10 Mbps = 1 hop

The maximum hop count is 15.

Anything more than 15 hops is considered unreachable, and RIP will not install the route.

Because of this limit:

- RIP cannot scale to large networks
- RIP is rarely used in real production networks
- It can still be used in:
 - Small networks
 - Lab environments
 - Simple IGP demonstrations

RIP Versions:

RIP has three versions

- RIPv1 – IPv4
- RIPv2 – IPv4
- RIPv6 – IPv6 (not covered here)

RIP Message Types

RIP uses two message types:

1. **Request**
 - Asks neighbouring RIP routers to send their routing table
2. **Response**
 - Sends the local router's routing table to neighbours

By default:

- RIP sends updates every 30 seconds

This can cause network congestion in networks with many routers.

RIPv1 vs RIPv2

RIPv1 Characteristics:

- Very old protocol
- Advertises only classful networks
- Does not support VLSM or CIDR
- Does not include subnet mask in updates

Subnet mask assumptions:

- Class A → /8
- Class B → /16
- Class C → /24

RIPv1 Forced Classful Examples:

Actual Subnet	Advertised As
10.1.1.0/24	10.0.0.0
172.16.192.0/18	172.16.0.0
192.168.1.4/30	192.168.1.0

This behaviour is not acceptable in modern networks.

RIPv1 messages are sent using broadcast:

- Destination: 255.255.255.255

RIPv2 Improvements:

- Supports VLSM and CIDR
- Includes subnet mask in route advertisements
- Advertises /30 as /30, not as classful
- Uses multicast, not broadcast

Multicast address used:

- 224.0.0.9

Broadcast vs Multicast:

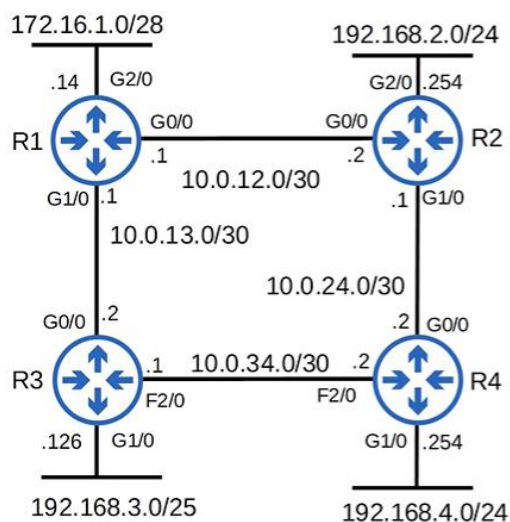
- Broadcast → all devices receive
- Multicast → only devices that joined the group receive

No deep multicast knowledge required at CCNA level.

RIP Basic Configuration (Introduction)

RIP configuration is not required for CCNA, but it is introduced because:

- Configuration is simple
- Mechanics are like OSPF
- Helps later with EIGRP and OSPF understanding



Assume all routers already run RIP.
We configure only R1.

RIP Configuration on R1

Enter RIP Mode:

R1(config)# router rip

```
R1(config)#router rip
```

Prompt changes to:

R1(config-router)#

Set Version 2:

```
R1(config)#router rip
R1(config-router)#version 2
```

Always use version 2.

Disable Auto-Summary:

```
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#no auto-summary
```

Auto-summary:

- Enabled by default
- Converts networks to classful
- Example:
172.16.1.0/28 → 172.16.0.0/16 (undesired)

Always use version 2 & no auto-summary

NETWORK Command Behaviour:

```
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#no auto-summary
R1(config-router)#network 10.0.0.0
```

Important facts:

- NETWORK command is classful
- 10.0.0.0 is treated as 10.0.0.0/8
- No subnet mask is required or accepted

What NETWORK Command Actually Does

The NETWORK command:

1. Finds interfaces with IPs in the specified range
2. Activates RIP on those interfaces
3. Forms adjacencies with RIP neighbours
4. Advertises the interface network prefix

It does not decide what network to advertise.

Step-by-Step: network 10.0.0.0

- Assumed range: 10.0.0.0/8
- Matches:
 - G0/0 → 10.0.12.1
 - G1/0 → 10.0.13.1

Result:

- RIP activated on G0/0 and G1/0
- Adjacencies formed with R2 and R3
- Advertised networks:
 - 10.0.12.0/30
 - 10.0.13.0/30

Not advertised:

- 10.0.0.0/8

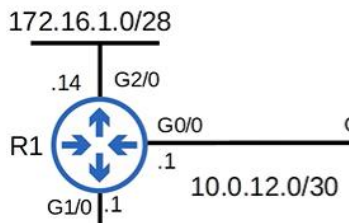
Step-by-Step: network 172.16.0.0

- Assumed range: 172.16.0.0/16
- Matches:
 - G2/0 → 172.16.1.14

Result:

- RIP activated on G2/0
- No neighbours on that interface
- Advertised network:
 - 172.16.1.0/28

Passive Interface



Even without neighbours, RIP still sends updates out G2/0.
This is unnecessary traffic.

Configure Passive Interface:

```
R1(config-router)#passive-interface g2/0
```

- Command is entered in RIP configuration mode
- Stops sending RIP updates out G2/0
- Still advertises 172.16.1.0/28 to neighbours

Recommended:

- Use passive-interface on all interfaces without RIP neighbours

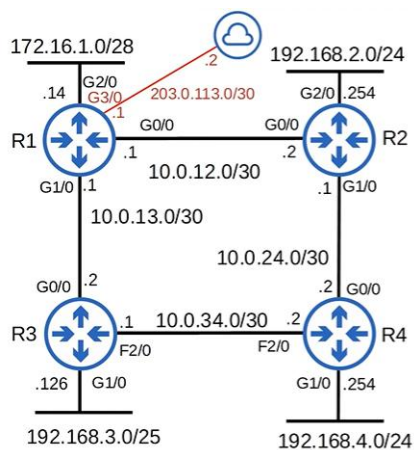
Same command exists in:

- EIGRP
- OSPF

Default Route Advertisement

An Internet connection is added to R1:

203.0.113.0/30 via G3/0



Default route configured:

```
R1(config)#ip route 0.0.0.0 0.0.0.0 203.0.113.2
```

Routing table shows:

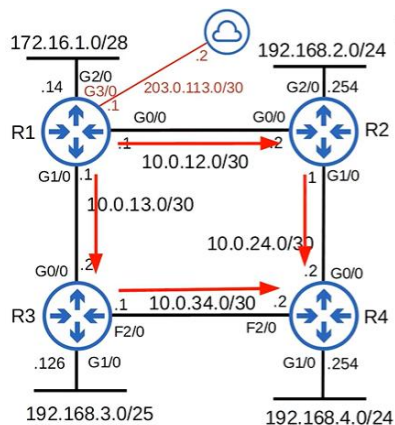
Gateway of last resort is 203.0.113.2 to network 0.0.0.0

```
Gateway of last resort is 203.0.113.2 to network 0.0.0.0
S* 0.0.0.0/0 [1/0] via 203.0.113.2
    10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C   10.0.12.0/30 is directly connected, GigabitEthernet0/0
L   10.0.12.1/32 is directly connected, GigabitEthernet0/0
```

Advertising Default Route via RIP

```
R1(config-router)#default-information originate
```

- Entered in RIP configuration mode
- R1 advertises default route to R2 and R3
- R2 and R3 advertise it to R4



R4 Routing Table Behaviour:

R4 shows: Gateway of last resort is 10.0.34.1 to network 0.0.0.0

```
R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l -
       + - replicated route, % - next hop override

Gateway of last resort is 10.0.34.1 to network 0.0.0.0

R* 0.0.0.0/0 [120/2] via 10.0.34.1, 00:00:06, FastEthernet2/0
    [120/2] via 10.0.24.1, 00:00:01, GigabitEthernet0/0
```


But two equal routes exist:

- Via R3 (Fast Ethernet)
- Via R2 (Gigabit Ethernet)

Because:

- Same hop count
- RIP treats all links equally

Result:

- Load balancing occurs

OSPF also uses default-information originate

SHOW IP PROTOCOLS

```
R1#show ip protocols
*** IP Routing is NSF aware ***
```

Shows:

- Routing protocol in use: RIP

```
Routing Protocol is "rip"
```

- Timers (not covered now)

```
Sending updates every 30 seconds, next due in 28 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
```

- Version: 2

```
Default version control: send version 2, receive version 2
Interface          Send Recv  Triggered RIP  Key-chain
GigabitEthernet0/0  2     2
GigabitEthernet1/0  2     2
```

- Auto-summary: disabled

```
Automatic network summarization is not in effect
```

- Maximum paths: 4 (default)

```
Maximum path: 4
```

Maximum Paths:

Default: Maximum paths: 4

Changeable with: maximum-paths 8

```
R1(config-router)#maximum-paths ?  
<1-32>  Number of paths  
  
R1(config-router)#maximum-paths 8
```

- Range: 1–32
- Controls ECMP load balancing
- Same command in:
 - RIP
 - EIGRP
 - OSPF

NETWORK Section Review:

Displayed networks:

- Only indicate which interfaces run RIP
- Not the advertised prefixes

```
Routing for Networks:  
  10.0.0.0  
 172.16.0.0
```

Passive interfaces:

- G2/0 listed

```
Passive Interface(s):  
GigabitEthernet2/0
```

Routing information sources:

- 10.0.12.2 (R2)
- 10.0.13.2 (R3)

```
Routing Information Sources:  
Gateway      Distance    Last Update  
10.0.12.2      120        00:00:21  
10.0.13.2      120        00:00:06
```

Administrative Distance:

Default RIP AD: 120

```
Distance: (default is 120)
```

Change with: distance 85

```
R1(config-router)#distance ?  
  <1-255> Administrative distance  
  
R1(config-router)#distance 85
```

- Range: 1–255
- Lower AD = preferred route
- Example:
 - RIP (85) preferred over EIGRP (90)

Same command applies to:

- EIGRP
- OSPF

End of RIP

Enhanced Interior Gateway Routing Protocol

Introduction to EIGRP

Let's move on to EIGRP.

You'll notice that many things are similar to RIP.

EIGRP stands for Enhanced Interior Gateway Routing Protocol.

It is an improved version of IGRP (Interior Gateway Routing Protocol).

EIGRP was originally Cisco proprietary.

Cisco later published it so other vendors could implement it, but not the entire protocol was opened, and very few vendors implemented it.

So in practice, EIGRP is still considered a Cisco-only protocol.

EIGRP Type and Capabilities

EIGRP is considered an advanced or hybrid distance vector routing protocol.

It improves on the basic behaviour of RIP:

- Much faster convergence
- No 15 hop-count limit
- Supports very large networks

EIGRP sends messages using the multicast address: 224.0.0.10

Comparison:

- RIPv1 → Broadcast (255.255.255.255)
- RIPv2 → Multicast (224.0.0.9)
- **EIGRP → Multicast (224.0.0.10)**

These multicast addresses are exam relevant.

Unequal-Cost Load Balancing

EIGRP is the only IGP that supports unequal-cost load balancing.

By default:

- Performs ECMP over 4 equal-cost paths, like RIP

But EIGRP can:

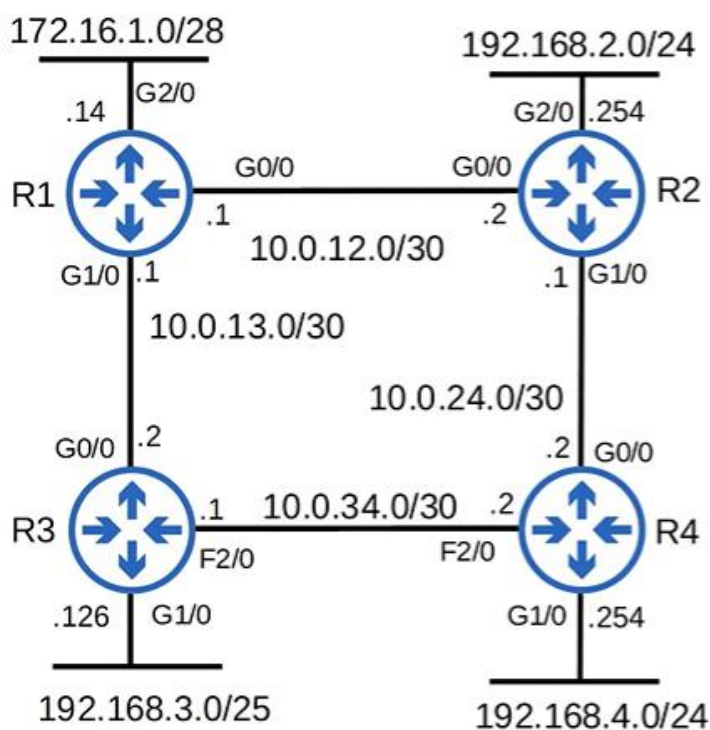
- Load-balance over unequal-cost paths
- Send more traffic on faster paths
- Send less traffic on slower paths

Traffic is distributed proportionally to bandwidth.

Despite its strengths:

- EIGRP is used less than OSPF
- Cisco focuses more on OSPF in the CCNA

EIGRP Basic Configuration (Introduction)



RIP is removed.

Only **EIGRP** is running.

Entering EIGRP Configuration Mode:

```
R1(config)#router eigrp 1
```

- 1 is the Autonomous System (AS) number
- AS numbers must match between routers
- R2, R3, and R4 are already using AS 1

If AS numbers don't match → no adjacency

Auto-Summary in EIGRP:

```
R1(config-router)#no auto-summary
```

Auto-summary:

- Works the same as in RIP
- Advertises classful networks
- Should be disabled

Depending on IOS version:

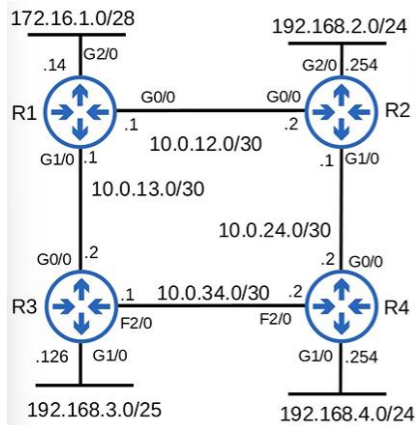
- It may be enabled or disabled by default
- Always verify and disable if enabled

Passive Interface in EIGRP

```
R1(config-router)#passive-interface g2/0
```

- Same command as RIP
- Prevents sending EIGRP messages out G2/0
- Still advertises the network to neighbours

EIGRP NETWORK Command (Without Mask):



```
R1(config-router)#network 10.0.0.0
```

If no mask is specified:

- EIGRP assumes classful
- 10.0.0.0 → /8

This command:

- Does not advertise 10.0.0.0/8
- Activates EIGRP on interfaces whose IP starts with 10

Result:

- G0/0 (10.0.12.1)
- G1/0 (10.0.13.1)

EIGRP is activated on both interfaces.

EIGRP NETWORK Command (With Wildcard Mask):

```
R1(config-router)#network 172.16.1.0 0.0.0.15
```

This activates EIGRP on G2/0.

If this feels confusing, that's expected.

This is your first exposure to wildcard masks.

What Is a Wildcard Mask?

EIGRP uses a wildcard mask, not a subnet mask.

A wildcard mask is an inverted subnet mask:

- Subnet mask 1 → Wildcard 0
- Subnet mask 0 → Wildcard 1

Subnet Mask → Wildcard Mask Examples:

11111111 . 11111111 . 11111111 . 00000000
255 . 255 . 255 . 0



00000000 . 00000000 . 00000000 . 11111111
0 . 0 . 0 . 255

/24

11111111 . 11111111 . 00000000 . 00000000
255 . 255 . 0 . 0



00000000 . 00000000 . 11111111 . 11111111
0 . 0 . 255 . 255

/16

11111111 . 00000000 . 00000000 . 00000000
255 . 0 . 0 . 0



00000000 . 11111111 . 11111111 . 11111111
0 . 255 . 255 . 255

/8

11111111 . 11111111 . 11111111 . 11110000
 255 . 255 . 255 . 240



00000000 . 00000000 . 00000000 . 00001111
 0 . 0 . 0 . 15

/28

11111111 . 11111111 . 11111111 . 100000
 255 . 255 . 255 . 128



00000000 . 00000000 . 00000000 . 01111111
 0 . 0 . 0 . 127

/25

Shortcut Method:

Subtract each subnet mask octet from **255**.

11111111 . 11111111 . 11111000 . 00000000
 255 . 255 . 248 . 0

255 - 255 ↓ 255 - 255 ↓ 255 - 248 ↓ 255 - 0 ↓

00000000 . 00000000 . 00000111 . 11111111
 0 . 0 . 7 . 255

/21

How Wildcard Masks Match Interfaces

Rules:

- 0 → bits must match
- 1 → bits don't need to match

Example 1 (Match)

R1 G2/0 IP address:			
10101100	. 00010000	. 00000001	. 00001110
172	. 16	. 1	. 14
EIGRP network command:			
10101100	. 00010000	. 00000001	. 00000000
172	. 16	. 1	. 0
00000000	. 00000000	. 00000000	. 00001111
0	. 0	. 0	. 15

- First 28 bits must match
- They do
- EIGRP activates

Example 2 (No Match)

R1 G2/0 IP address:			
10101100	. 00010000	. 00000001	. 00001110
172	. 16	. 1	. 14
EIGRP network command:			
10101100	. 00010000	. 00000001	. 00000000
172	. 16	. 1	. 0
00000000	. 00000000	. 00000000	. 00000111
0	. 0	. 0	. 7

- First 29 bits must match
- One bit does not
- No activation

Example 3 (Match):

R1 G2/0 IP address:			
10101100	. 00010000	. 00000001	. 00001110
172	. 16	. 1	. 14
EIGRP network command:			
10101100	. 00010000	. 00000001	. 00001000
172	. 16	. 1	. 8
00000000	. 00000000	. 00000000	. 00000111
0	. 0	. 0	. 7

- First 29 bits match
- EIGRP activates

Example 4 (Large Match):

R1 G2/0 IP address:			
10101100	. 00010000	. 00000001	. 00001110
172	. 16	. 1	. 14
EIGRP network command:			
10101000	. 00000000	. 00000000	. 00000000
168	. 0	. 0	. 0
00000111	. 11111111	. 11111111	. 11111111
7	. 255	. 255	. 255

- Only first 5 bits must match
- They do
- EIGRP activates

Notes on Using Wildcard Masks

- You can use many wildcard masks
- Usually best practice:
 - Match the interface prefix length
 - Or use /32 → 0.0.0.0 to match exact IP

/32 wildcard mask: 0.0.0.0

Important reminder:

- NETWORK command only activates EIGRP
- Router advertises the interface's actual network prefix
- Example: 172.16.1.0/28

SHOW IP PROTOCOLS (EIGRP)

```
R1#show ip protocols
*** IP Routing is NSF aware ***
```

Key points:

- Routing protocol: EIGRP 1
- Metric uses bandwidth and delay
 - K1 = 1 (bandwidth)
 - K3 = 1 (delay)
- Metric calculation:
 - Lowest bandwidth in path
 - Sum of all delays
- K2, K4, K5 = 0 (not used)

```
and the formula is:
Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
```

Router ID in EIGRP:

Router ID selection order:

1. Manually configured router ID
2. Highest loopback IP
3. Highest physical interface IP

On R1:

- Highest IP = 172.16.1.14
- So router ID = 172.16.1.14

```
Router-ID: 172.16.1.14
```

Router ID:

- Not a real IP
- Just a 32-bit number

Configure Router ID:

```
R1(config-router)#eigrp router-id ?  
A.B.C.D  EIGRP Router-ID in IP address format  
  
R1(config-router)#eigrp router-id 1.1.1.1
```

Manual configuration always wins.

```
Router-ID: 1.1.1.1
```

Remaining SHOW IP PROTOCOLS Fields

- Auto-summarization: disabled
- Maximum paths: 4

```
Automatic Summarization: disabled  
Maximum path: 4
```

- Routing for networks:
 - 10.0.0.0
 - 172.16.1.0/28

```
Routing for Networks:  
10.0.0.0  
172.16.1.0/28
```

- Passive interface: G2/0

```
Passive Interface(s):
  GigabitEthernet2/0
```

- Neighbours: R2 and R3

```
Routing Information Sources:
  Gateway         Distance      Last Update
  10.0.12.2        90            00:00:23
  10.0.13.2        90            00:00:23
```

- Administrative Distance:

- Internal: **90**
- External: **170**

```
Distance: internal 90 external 170
```

Remember both values.

Show ip protocols:

```
R1#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "eigrp 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  EIGRP-IPv4 Protocol for AS(1)
    Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
    NSF-aware route hold timer is 240
    Router-ID: 172.16.1.14
    Topology : 0 (base)
    Active Timer: 3 min
    Distance: internal 90 external 170
    Maximum path: 4
    Maximum hopcount 100
    Maximum metric variance 1

  Automatic Summarization: disabled
  Maximum path: 4
  Routing for Networks:
    10.0.0.0
    172.16.1.0/28
  Passive Interface(s):
    GigabitEthernet2/0
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.0.12.2        90            00:00:23
    10.0.13.2        90            00:00:23
  Distance: internal 90 external 170
```

EIGRP in Show Routing Table

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C    10.0.12.0/30 is directly connected, GigabitEthernet0/0
L    10.0.12.1/32 is directly connected, GigabitEthernet0/0
C    10.0.13.0/30 is directly connected, GigabitEthernet1/0
L    10.0.13.1/32 is directly connected, GigabitEthernet1/0
D    10.0.24.0/30 [90/3072] via 10.0.12.2, 00:11:09, GigabitEthernet0/0
D    10.0.34.0/30 [90/28416] via 10.0.13.2, 00:11:09, GigabitEthernet1/0
172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.16.1.0/28 is directly connected, GigabitEthernet2/0
L    172.16.1.14/32 is directly connected, GigabitEthernet2/0
D    192.168.2.0/24 [90/3072] via 10.0.12.2, 00:11:09, GigabitEthernet0/0
192.168.3.0/25 is subnetted, 1 subnets
D    192.168.3.0 [90/3072] via 10.0.13.2, 00:11:10, GigabitEthernet1/0
D    192.168.4.0/24 [90/3328] via 10.0.12.2, 00:11:09, GigabitEthernet0/0
```

D 10.0.24.0/30 [90/3072]

D 10.0.34.0/30 [90/28416]

D 192.168.4.0/24 [90/3328]

Notes:

- EIGRP routes marked with D
- Metrics are large numbers
- Harder to interpret than RIP or OSPF
- In large networks, metrics grow much larger

Final Review

- Covered **RIP basics and configuration**
- Covered **EIGRP basics and configuration**
- Learned **wildcard masks**
- Learned **EIGRP metrics and router ID**
- This knowledge makes **OSPF much easier**

Unequal-cost load balancing exists in EIGRP,
but it will be covered in the lab video.